MathDB 367325: polynomial blocks for every Motzkin convolution power

Principia Math In Progress

18 September 2026

Task 1: Full proof of the determinant block pattern

Let $M(z)=\sum_{n\ge 0}M_n z^n$ be the Motzkin generating function, so $M=1+zM+z^2M^2$. For an integer $r\ge 1$, write

$$M(z)^r=\sum_{n\ge 0}a_n^{(r)}z^n,\qquad H_N^{(r)}=\det(a_{i+j}^{(r)})_{0\le i,j<N},\qquad H_0^{(r)}=1.$$

Wang and Zhang's Conjecture 8 (arXiv:2502.21050, p. 4) predicts shifted block lengths r when $3\mid r$ and $3r$ otherwise. We establish the determinant-level assertion in the precise form below: each residue class is a polynomial in the block index, with an explicitly determined parity factor. These are the polynomial patterns displayed in their Theorems 1, 2, 4–6. The block length here is not the number of continued-fraction transformations: for example the source uses two transformations for block length three and ten for block length twelve. No minimality of a block length or periodicity of the numerical values themselves is required.

Theorem 1. *Set*

$$p=\operatorname{lcm}(r,3)=\begin{cases}r, & 3\mid r,\\ 3r, & 3\nmid r,\end{cases}\qquad \epsilon=(-1)^{\binom{r}{2}},\qquad \kappa=\begin{cases}r, & 3\mid r,\\ r-1, & 3\nmid r,\end{cases}\qquad D=\binom{\kappa}{2}.$$

For each $s\in\{0,\dots,p-1\}$ there is an integer-valued polynomial $B_{r,s}(X)\in\mathbb{Q}[X]$, of degree at most D, such that

$$H_{pn+s}^{(r)}=\epsilon^n B_{r,s}(n)\qquad (n\ge 0). \tag{1}$$

The proof supplies an exact determinant of order $2r-2$ for every $H_N^{(r)}$, as well as a finite formula for every block polynomial. The convention is $D=0$ when $r=1$.

Define monic shifted Chebyshev polynomials by

$$P_{-1}(t)=0,\quad P_0(t)=1,\quad P_1(t)=t-1,\qquad P_{j+1}(t)=(t-1)P_j(t)-P_{j-1}(t).$$

Thus $P_j(1+2\cos\theta)=\sin((j+1)\theta)/\sin\theta$. Let μ be the probability measure

$$d\mu(t)=\frac{\sqrt{4-(t-1)^2}}{2\pi}\,dt\quad(-1\le t\le 3).$$

The substitution $t=1+2\cos\theta$ gives

$$\int P_i(t)P_j(t)\,d\mu(t)=\frac{2}{\pi}\int_0^{\pi}\sin((i+1)\theta)\sin((j+1)\theta)\,d\theta=\delta_{ij}. \tag{2}$$

Its moments are M_n: expansion of $(1+2\cos\theta)^n$ gives

$$\int t^n\,d\mu(t)=\sum_{k=0}^{\lfloor n/2\rfloor}\binom{n}{2k}\frac{1}{k+1}\binom{2k}{k}=M_n.$$

Here the odd centered moments vanish by symmetry and the even ones are the displayed Catalan numbers, by the elementary sine integral.

Lemma 1 (The weight for a convolution power)**.** *For $Q_r(t) = t^{r-1}P_{r-1}(t)$ and every $n \geq 0$,*

$$a_n^{(r)} = \int t^n Q_r(t)\, d\mu(t). \tag{3}$$

Proof. In formal Laurent series at infinity, put $G(w) = w^{-1}M(w^{-1}) = \sum_{n\geq 0} M_n w^{-n-1}$. The Motzkin equation gives $G^2 = (w-1)G - 1$. Induction using the recurrence for P_j consequently gives

$$G(w)^r = P_{r-1}(w)G(w) - P_{r-2}(w).$$

Multiply by w^{r-1}. The left side is $w^{-1}M(w^{-1})^r$. The second term on the right is a polynomial, and the coefficient of w^{-n-1} in the first is $\sum_j [t^j]Q_r(t)\, M_{n+j}$, proving (3). □

We recall and prove the needed form of the classical Christoffel determinant identity. It is also Theorem 1 of Krattenthaler, arXiv:2101.04225, with confluent evaluation as in its Proposition 5; the identity itself is an established tool.

Lemma 2 (Polynomial modification of moments)**.** *Let $Q(t) = \prod_{i=1}^{d}(t-\alpha_i)$ be monic and let the monic P_j be orthonormal for μ. For distinct roots,*

$$\det\left(\int t^{i+j}Q(t)\, d\mu(t)\right)_{0\leq i,j<N} = (-1)^{Nd}\frac{\det(P_{N+j}(\alpha_i))_{1\leq i\leq d,\, 0\leq j<d}}{\det(\alpha_i^j)_{1\leq i\leq d,\, 0\leq j<d}}. \tag{4}$$

Repeated roots are handled by replacing the rows at a root of multiplicity m by derivative evaluations of orders $0, \ldots, m-1$, each divided by the factorial of its order, in both determinants. The confluent denominator is nonzero.

Proof. Expansion of two determinants and integration term by term give the left side as

$$\frac{1}{N!}\int\cdots\int \Delta(x_1,\ldots,x_N)^2 \prod_{k=1}^{N} Q(x_k) \prod_{k=1}^{N} d\mu(x_k),$$

where Δ uses increasing powers in its columns. The mixed Vandermonde, with the x variables followed by the α variables, satisfies

$$\Delta(x,\alpha) = \Delta(x)\Delta(\alpha)\prod_{k,i}(\alpha_i - x_k).$$

Replace its monomials, and those in the remaining $\Delta(x)$, by the monic P_j, which does not change either determinant. Expand the mixed determinant along its last d rows. By orthonormality, integration against the other determinant kills every term except that using columns $0, \ldots, N-1$ for the first N rows. That term has positive expansion sign and integral $N!$. The remaining minor is the numerator in (4), and $\prod Q(x_k)$ contributes $(-1)^{Nd}$.

For colliding roots, divide numerator and denominator by their common within-cluster Vandermonde factors and take limits. Taylor expansion gives exactly the stated derivative rows. These derivative evaluations are linearly independent on polynomials of degree less than d: a polynomial in their common kernel is divisible by Q and hence is zero. The confluent denominator is therefore nonzero. Both sides depend continuously on the polynomial coefficients, so the identity remains valid. The empty-determinant case is included. □

Proof of Theorem 1. For $r = 1$, (2) gives $H_N^{(1)} = 1$, which proves the assertion. Suppose $r \geq 2$ and put $d = 2r-2$. The roots of P_{r-1} are simple and are

$$\rho_j = 1 + 2\cos(j\pi/r), \qquad 1 \leq j \leq r-1.$$

One equals zero exactly when $3 \mid r$, in which case its index is $j_0 = 2r/3$, an even integer. Thus Q_r has a root zero of multiplicity κ, together with the nonzero ρ_j, all simple.

Let $\mathcal{E}_N$ be the $d\times d$ matrix whose columns are $P_N, P_{N+1}, \dots, P_{N+d-1}$ and whose rows apply the following functionals: $f \mapsto f^{(k)}(0)/k!$ for $0 \le k < \kappa$, then $f \mapsto f(\rho_j)$ for each nonzero ρ_j, in any fixed order. Let $\mathcal{V}$ use the same rows and columns $1, t, \dots, t^{d-1}$. Lemmas 1 and 2, and evenness of d, give

$$H_N^{(r)} = \frac{\det \mathcal{E}_N}{\det \mathcal{V}}. \tag{5}$$

In particular no division by a possibly vanishing $H_N^{(r)}$ occurs.

Set $N = pn + s$. The integer p/r is odd. For a nonzero ρ_j, the sine formula gives, for every column offset h,

$$P_{pn+s+h}(\rho_j) = (-1)^{nj} P_{s+h}(\rho_j).$$

Factor this sign out of each such row. Their product is $(-1)^{n\binom{r}{2}} = \epsilon^n$: the only possibly omitted index is the even integer j_0.

For the zero rows use $t = 1 + 2\cos\theta$ near $\theta_0 = 2\pi/3$. Differentiation in t acts as $-(2\sin\theta)^{-1} d/d\theta$. After k differentiations of $\sin((L+1)\theta)/\sin\theta$, the value at θ_0 is a linear combination of

$$(L+1)^a \sin((L+1)\theta_0), \qquad (L+1)^a \cos((L+1)\theta_0), \qquad 0 \le a \le k,$$

with coefficients independent of L. This follows directly by induction with the product rule; $\sin\theta_0 \neq 0$ ensures all coefficients are finite. Since $3 \mid p$, substituting $L = pn + s + h$ makes the sine and cosine factors independent of n. Each entry in the derivative row of order k is therefore a polynomial in n of degree at most k.

After removing the simple-root signs, the determinant in (5) is a polynomial in n of degree at most $0 + 1 + \cdots + (\kappa - 1) = D$. Its denominator is a fixed nonzero constant, so this proves (1) initially over $\mathbb{R}$. Its values at all nonnegative integers are integers, because both $H_N^{(r)}$ and ϵ are integers. Newton interpolation now gives the explicit expression

$$B_{r,s}(X) = \sum_{j=0}^{D} \binom{X}{j} \sum_{h=0}^{j} (-1)^{j-h} \binom{j}{h} \epsilon^h H_{ph+s}^{(r)}. \tag{6}$$

Hence $B_{r,s} \in \mathbb{Q}[X]$ and is integer-valued (also at negative integers).

For completeness, (5) has a version requiring only integer polynomial arithmetic. Reduce P_{N+j} modulo the monic polynomial Q_r and write its remainder as $\sum_{i=0}^{d-1} c_{ij} t^i$. Every evaluation functional defining $\mathcal{E}_N$ annihilates multiples of Q_r, so $\mathcal{E}_N = \mathcal{V}(c_{ij})$ and

$$H_N^{(r)} = \det(c_{ij})_{0 \le i,j < 2r-2}. \tag{7}$$

This proves the promised finite determinant formula and completes the proof for every exponent, residue, and nonnegative block index. □

Corollary 1. *The determinant sequence satisfies $(E^p - \epsilon)^{D+1} H^{(r)} = 0$, where $(EH)_N = H_{N+1}$. In particular $\sum_{N \ge 0} H_N^{(r)} z^N$ is rational, with denominator dividing $(1 - \epsilon z^p)^{D+1}$.*

Proof. On each residue class, applying $E^p - \epsilon$ takes a polynomial to ϵ times its forward difference. After $D+1$ applications it vanishes. Multiplication of the ordinary generating series by the displayed denominator therefore leaves a polynomial. □

Attribution and scope. The source authors conjectured the block lengths and provided explicit cases. Polynomial modifications of orthogonal-polynomial moments and their Christoffel determinant formula are classical; the latter is proved here for completeness and attributed above. The contribution here is the convolution weight Q_r, the analysis of its roots, and the resulting uniform signed polynomial blocks. This addresses Conjecture 8 for the determinants; it does not assert the separate numerical formulas of Conjectures 3 and 7 or a specific orbit of the continued-fraction transformation. The displayed sign is derived independently, rather than imported from those formulas. Exactly one MathDB problem is addressed. The following tasks are proposed directions, not additional solved problems.

Task 2: Find sharp degrees and leading coefficients

Determine the exact degree of every nonzero $B_{r,s}$, classify the identically zero residues, and evaluate the leading coefficients. The general bound $\binom{\kappa}{2}$ need not be sharp.

Task 3: Evaluate distinguished residue classes

Use the fixed-size determinant to prove explicit formulas for $B_{r,0}$, $B_{r,1}$, and the paired residues in the source's Conjectures 3 and 7, including their correct sign factors.

Task 4: Determine the least block length

Classify all cancellations that allow a smaller signed polynomial block length or a smaller generating-function denominator than those supplied by the theorem.

Task 5: Extend to weighted Motzkin moments

Replace the horizontal-step weight one by a real parameter. Determine which parameters produce polynomial blocks, and describe the exponential-polynomial behavior when the relevant root angles are not rational multiples of π.